INCH-POUND

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SUPERSEDING
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19 March 1993

DETAIL SPECIFICATION

# THRUSTER, CARTRIDGE ACTUATED, M1A2 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for the M1A2 Cartridge Actuated Thruster Metal Parts Assembly (see 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SURFACE WARFARE CENTER, INDIAN HEAD DRAWINGS (CAGE CODES 19200 AND 19203)

8593855 - Thruster, Cartridge Actuated, M1A2 Metal Parts Assembly

(Copies of this document are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 3601-3 - Fluid power systems-O-rings-Part 3: Quality acceptance criteria

(Copies of this document are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. All materials used in the manufacture of the thrusters shall conform to Drawing 8593855. Any departures from the drawing shall require written approval by the Technical Authority (see 6.4.2) prior to manufacture, as specified (see 6.2).

3.2.1 Raw material. The raw material shall be selected from the classification list contained in the raw material specification, located in Drawing 8593855, based on the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hydrostatic pressure. The body and head shall withstand a hydrostatic pressure of 7,500 pounds per square inch for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.4.1).

3.4 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 48±4 pounds is applied at the steady rate of 0.125 inch per minute.

3.5 Metal parts assembly. Parts shall be combined to produce an assembly that conforms to the requirements of the metal parts assembly Drawing 8593855. All threaded joints shall be assembled handtight.

3.6 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.7 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than 5 degrees in either direction using the procedures in 4.6.4.

3.8 Workmanship. Thrusters shall be free of cracks, splits, or other defects that might prevent proper mating with connecting parts or proper functioning of the device.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr that might interfere with the assembly or function of the item or might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77 °F).

b. Altitude: local altitude.

c. Humidity: 50 percent relative, maximum.

d. Atmospheric pressure: local pressure.



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4.3 First article inspection. Unless otherwise specified (see 6.2), a first article sample of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish, and 50 shear pins from the same lot of shear pins tested shall be subjected to first article testing. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as those to be used for regular production and shall be obtained from the same source of supply. The inspection shall include the examination of 4.5 and the tests of 4.6 to ensure compliance with Drawing 8593855. Failure of any part to meet the requirements of the drawings or specifications shall be cause for rejection of the first article lot represented.

4.4 Conformance inspection. Conformance inspection shall include the examination of 4.5 and the tests of 4.6.

4.4.1 Lot. Unless otherwise specified (see 6.2), formation and identification of lots shall be in accordance with ASQ/ANSI Z1.4.

4.5 Examination. One hundred percent examination shall be performed for all critical defects on Drawing 8593855. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects on Drawing 8593855. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with ISO 3601-3. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.6 Inspection testing. The following tests shall be performed:

4.6.1 Hydrostatic pressure. Prior to application of protective finish, the body and end cap shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.3. Each body and end cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with 3.3 shall be cause for rejection of the subassembly.

4.6.2 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.4. The test shall be performed in a fixture in conjunction with a tensile testing machine. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.4 shall be cause for rejection of the lot.

4.6.3 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.6.4 Residual magnetism. The assembly shall be tested for excessive residual magnetism using a compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90-degree rotation of the assembly for four full passes. Each assembly shall be subjected to the test for excessive residual magnetism as specified. Failure of any unit in the inspection sample to comply with 3.7 shall be cause for rejection of the assembly.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)



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6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems. The thruster assembly is a human-rated, aviation critical safety item with no commercial equivalent or substitute.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1 and 6.3).

c. Requirement for Technical Authority approval for departures from drawings (see 3.2).

d. Any special instruction requirements (see 4.2).

e. First article sample size, if other than as specified (see 4.3).

f. Formation and identification of lots if other than specified (see 4.4.1).

g. Packaging requirements (see 5.1). Recommended packaging requirement is per SPI F00-891-5488.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.4.2 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.5 Subject term (key word) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

## CONCLUDING MATERIAL

Custodians:
Navy – OS
Air Force – 70

Preparing activity:
Navy – OS
(Project 1377-2024-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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